Oxalate

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Identity Sheet

Property	Value
Name	Oxalate
Other names	Oxalic acid, ethanedioic acid, C2O4(2-)
Chemical formula	$C_2H_2O_4$ (acid); $C_2O_4^{2-}$ (dianion)
Molecular weight	90.03 g/mol
CAS number	144-62-7
PubChem CID	971
pKa	pKa1 = 1.25; pKa2 = 4.27
Solubility	Sodium oxalate: ~37 g/L; Calcium oxalate: 0.0067 g/L (very insoluble)

Structural formula:

```
O=C-C=O
| |
OH OH (oxalic acid, fully protonated)
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Oxalate is the simplest dicarboxylic acid. At urinary pH, it exists primarily as the **oxalate dianion ($C_2O_4^{2-}$)**. Its clinical importance lies in its propensity to form **calcium oxalate (CaOx) crystals** – the most common component of kidney stones (~75-80% of all stones). Even small increases in urinary oxalate dramatically raise CaOx supersaturation.

Molecules Not to Be Confused With

Molecule	Formula	MW (g/mol)	Key Difference
Citrate	$C_6H_5O_7^{3-}$	189.10	Stone inhibitor (chelates Ca); opposite role from oxalate
Glycolate	$C_2H_3O_3^-$	75.04	Precursor of oxalate; elevated in primary hyperoxaluria type 1
Glyoxylate	$C_2HO_3^-$	73.03	Key intermediate in oxalate synthesis; substrate of AGT enzyme

Medical Information

Origin

Endogenous ~50% of urinary oxalate is endogenously produced, primarily in the liver. Pathways: glyoxylate -> oxalate (via lactate dehydrogenase or glycolate oxidase); ascorbic acid (vitamin C) -> oxalate (non-enzymatic degradation); hydroxyproline -> oxalate.

Exogenous ~40-50% from dietary sources: spinach, rhubarb, beetroot, nuts, chocolate, tea, sweet potatoes. Vitamin C supplements (>1 g/day) increase oxalate via non-enzymatic degradation.

Primary & Secondary Biological Roles

Primary role: - **Metabolic waste product:** Oxalate has no known physiological function in humans. It is purely a waste product of glyoxylate and ascorbate metabolism.

Secondary roles: - **Stone formation:** Calcium oxalate precipitation is the basis of most kidney stones. - **Plant defence:** In plants, oxalate crystals deter herbivory (raphides). Not relevant to human physiology.

Catabolism and Elimination Pathway

- Humans cannot metabolise oxalate (no oxalate-degrading enzyme, unlike gut bacteria Oxalobacter formigenes).
- **Renal excretion** is the sole elimination route: filtered at glomerulus + secreted by proximal tubule (SLC26A6 transporter).
- Intestinal Oxalobacter formigenes degrades dietary oxalate before absorption; loss of this bacterium (antibiotics) increases oxalate absorption.

Expression in Humans

Normal Levels

Compartment	Reference Range
Plasma oxalate	1-5 umol/L
Urinary oxalate (24-h)	<40-45 mg/day (<0.5 mmol/day)
Urinary oxalate concentration	10-40 mg/L

Factors Influencing Levels **Hyperoxaluria (>45 mg/day):** - Primary hyperoxaluria (types 1, 2, 3 – genetic; severe; oxalate >100 mg/day) - Enteric hyperoxaluria (fat malabsorption: Crohn's, bariatric surgery) - Dietary hyperoxaluria (high-oxalate foods, vitamin C excess) - Ethylene glycol poisoning (metabolised to oxalate)

Low urinary oxalate: - Low dietary oxalate - Pyridoxine (B6) supplementation (cofactor for AGT, reduces glyoxylate -> oxalate)

Associated Pathologies

Condition	Oxalate Pattern	Key Symptoms
CaOx nephrolithiasis	Mild hyperoxaluria (40-60 mg/day)	Renal colic, haematuria; 75-80% of all stones
Primary hyperoxaluria type 1	>100 mg/day; AGT deficiency	Recurrent stones, nephrocalcinosis, ESRD; ~1:120,000
Enteric hyperoxaluria	60-100 mg/day	Post-bariatric surgery, Crohn's; increased stone risk
Ethylene glycol poisoning	Massive oxaluria + CaOx crystals	Metabolic acidosis, AKI, CNS depression

Presence in Urine

Should it be normally present? Yes – oxalate is a normal urinary solute (10-40 mg/day).

Normal urinary levels: <40-45 mg/day.

Form in urine: Oxalate dianion ($C_2O_4^{2-}$) at urinary pH >4.3. Forms insoluble calcium oxalate crystals (monohydrate: whewellite; dihydrate: weddellite) when supersaturated.

Solubility: Calcium oxalate is extremely insoluble (Ksp ~2.3×10^{-9}). This is why even moderate increases in urinary oxalate dramatically increase stone risk. Sodium/potassium oxalate are soluble.

Detection in Urine

Available Clinical Assays

1. **Oxalate oxidase enzymatic assay:**
   - **Principle:** Oxalate oxidase converts oxalate + O2 -> 2CO2 + H2O2. H2O2 detected by peroxidase + chromogen (MBTH/DMAB).
   - **Detection:** 590 nm.
   - **LOD:** ~0.05 mg/dL (~5 umol/L).
   - **Advantages:** Specific for oxalate; automated.
2. **Ion chromatography:**
   - **Principle:** Anion exchange separation + conductivity detection.
   - **Detection:** Conductivity.
   - **LOD:** ~0.5 umol/L.
   - **Advantages:** Reference method; multi-anion.
3. **Capillary electrophoresis:**
   - **Principle:** Electrophoretic separation + UV detection at 200 nm.
   - **LOD:** ~1 umol/L.

Optimal Urine Type for Measurement

- **24-hour urine** acidified with HCl (6N, ~20 mL per 24-h container) to prevent CaOx crystallisation.
- Without acidification, oxalate precipitates as CaOx crystals -> falsely low result.
- Refrigerate during collection.

Actual Gold Standard

Oxalate oxidase enzymatic assay on automated platforms (Trinity Biotech, Roche). IC is the reference for research.

Interferences in Measurement

Interference	Effect	Mechanism
Ascorbic acid	Positive bias	Non-enzymatic conversion to oxalate during storage; also false elevation in some assays
CaOx crystal precipitation	Negative bias	Oxalate lost to crystals if urine not acidified
Glycolate (high)	Minor cross-reactivity	Some oxalate oxidase preparations

Research Detection Methods

Spectroscopy Detection (UV-Vis / NIR)

- Oxalate has weak UV absorption (<210 nm). Direct UV not practical in urine.
- Enzymatic colorimetric (590 nm): see clinical assay.
- NIR: No useful absorption.

Fluorescence Detection

- Oxalate is non-fluorescent. Indirect:
- **Enzymatic H2O2 -> Amplex Red (Ex 571 / Em 585 nm).** LOD: ~1 umol/L.
- **Ce(III) fluorescence quenching:** Oxalate quenches Ce3+ fluorescence (Ex 256 / Em 360 nm). LOD: ~5 umol/L.

Raman Detection

Peak (cm-1)	Assignment
~1460	C-O symmetric stretch
~1620	C=O asymmetric stretch
~890	C-C stretch

- Conventional Raman: LOD ~1 mM. Marginally adequate for hyperoxaluric urine.
- CaOx crystals have distinct Raman signatures used for stone composition analysis.

FTIR Detection

Band (cm-1)	Assignment	Notes
~1620	C=O asymmetric stretch	Strong
~1320	C-O symmetric stretch	–
~780	O-C-O bending	–

- ATR-FTIR can identify CaOx crystals in urine sediment (stone analysis). Not practical for dissolved oxalate quantification.

Voltammetry Detection

- Oxalate is oxidised at **+0.8-1.2 V vs Ag/AgCl** on carbon or platinum electrodes.
- LOD: ~1-10 umol/L on modified electrodes (PbO2, boron-doped diamond).
- Interference from ascorbic acid at similar potentials.

Other Detection Technologies

1. **GC-MS (after derivatisation):** LOD ~0.1 umol/L. Definitive.
2. **LC-MS/MS:** LOD ~0.5 umol/L without derivatisation.
3. **Paper-based oxalate oxidase assay:** Enzyme on paper + colour indicator. LOD: ~10 umol/L.

Summary Table: Detection Methods Comparison

Method	LOD	Key Parameter	Sample Prep	Strengths
Oxalate oxidase	~5 umol/L	590 nm	Acidified urine	Standard clinical

Method	LOD	Key Parameter	Sample Prep	Strengths
Ion chromatography	~0.5 umol/L	Conductivity	Dilution	Reference
Amplex Red fluorescence	~1 umol/L	Ex 571/Em 585 nm	Enzyme kit	Sensitive
Voltammetry (BDD)	~1 umol/L	+1.0 V	Buffer	Real-time
Raman (CaOx crystals)	N/A	1460 cm-1	Crystal isolation	Stone analysis
LC-MS/MS	~0.5 umol/L	m/z 89->61	Extraction	Definitive

References

1. StatPearls - Hyperoxaluria. https://www.ncbi.nlm.nih.gov/books/NBK551653/
2. KDIGO - Kidney Stone Guidelines.
3. PubChem - Oxalic acid, CID 971. https://pubchem.ncbi.nlm.nih.gov/compound/971
4. Trinity Biotech - Oxalate Assay Kit documentation.